Legislative Document Monitor

Executive Summary — Project Scope & Architecture

March 26, 2026

Project Overview

The Legislative Document Monitor is a web application designed to track, analyze, and respond to legislative activity across all 50 U.S. state legislatures and the U.S. Congress. The system provides real-time monitoring of bills and resolutions, automatic detection of document changes through diff comparison, and a suite of civic engagement tools for annotation, markup, and testimony preparation.

> **Mission:** Empower users to stay informed on legislative changes, organize their analysis, and produce actionable feedback for legislators — all through deterministic, non-AI-dependent processing.

Core Capabilities

- **Bill Monitoring & Diffing** — Ingest bill text from official APIs, store versioned snapshots, and compute line-by-line diffs to instantly surface what changed between versions.
- **Watchlists & Position Tracking** — Flag specific bills with a position stance (Support, Oppose, Watch, Neutral) and receive alerts when they're updated.
- **Keyword Monitoring** — Define terms to watch for across all ingested legislation. Automatic matching with contextual snippets.
- **Inline Annotations & Comment Chains** — Select specific language in bill text and attach threaded discussion comments for analysis and review.
- **Legislative Markup & Redlining** — Highlight, strikethrough, and suggest alternative language directly on bill text. Export as formatted documents for legislator feedback.
- **Testimony Drafting** — Structured workspace for composing written testimony linked to specific bill sections.
- **Legislative Calendar** — Track hearing dates, committee schedules, and floor votes.
- **Legislator Directory** — Searchable sponsor database with contact information and sponsored legislation.

- **Email Digests** — Configurable daily or weekly email summaries of changes, matches, and upcoming events.
- **Cross-Bill Comparison** — Side-by-side comparison of different bills from any jurisdictions.
- **Shareable Views** — Generate read-only links to annotated bill views for coordination with others.

Data Sources

Source	Coverage	Cost	Data Format
LegiScan API (Primary)	All 50 states + U.S. Congress	Free tier: 30,000 queries/month	Structured JSON
Congress.gov API (Secondary)	U.S. Congress (federal only)	Free with API key	JSON / XML

API keys are managed through an in-app Settings page and stored securely in the database. No AI or machine learning services are used — all processing (diffing, keyword matching, alerting) is handled by deterministic algorithms.

Technology Stack

Frontend
Vite + React

Backend
Express.js (Node.js)

Database
PostgreSQL

Migrations
Knex.js

Scheduling
node-cron

Diffing
diff (npm)

Email
Nodemailer (SMTP)

PDF Export
PDFKit / Puppeteer

Development Roadmap

Phase	Focus	Key Deliverables
Phase 1	**Foundation (MVP)**	Bill ingestion, diff engine, watchlists, dashboard, settings page, cron scheduler
Phase 2	**Engagement**	Inline annotations, comment chains, tags, collections, keyword matching, alerts system
Phase 3	**Advocacy**	Markup/redlining tools, PDF export, testimony workspace, status pipeline, Congress.gov API
Phase 4	**Awareness**	Legislative calendar, email digest service, legislator directory, bill relationship mapping
Phase 5	**Collaboration**	Cross-bill comparison, shareable links, multi-user support, responsive polish

Architecture Summary

The system follows a standard client-server architecture. The React frontend communicates with the Express.js backend via REST API. The backend manages a PostgreSQL database containing 22 tables spanning bill data, version history, diffs, user annotations, markups, testimony drafts, legislator records, calendar events, and configuration settings.

A built-in cron scheduler periodically polls the LegiScan and Congress.gov APIs to ingest new bills, detect version changes, compute diffs, and trigger alerts. An email digest service delivers configurable summaries via SMTP.

Key Design Principles

- **No AI Dependency** — All processing is deterministic. Diffing uses standard text comparison algorithms; keyword matching uses exact and pattern-based search; alerting follows rule-based triggers.
- **API-First Data** — Relies on structured API data from official sources (LegiScan, Congress.gov) rather than fragile web scraping.
- **Self-Hosted & Configurable** — API keys, polling intervals, and email settings are managed through the app's Settings page. The instance is fully self-contained.
- **Civic Engagement Focus** — Goes beyond passive monitoring with annotation, markup, testimony, and export tools designed for real legislative advocacy workflows.